

שאלה 1

2007 מועד א

חלק א

תכנון

--מבחן 2007 מועד א שאלה 6

-- תכנון רכיב alu

```
library IEEE
;use IEEE.STD_LOGIC_1164.ALL
;use IEEE.NUMERIC_STD.ALL

entity ALU is
    Port
        A : in unsigned(3 downto 0) -- קלט ראשון (4 ביטים)
        B : in unsigned(3 downto 0) -- קלט שני (4 ביטים)
        Opcode : in std_logic_vector(1 downto 0); -- Opcode (2 ביטים)
        Y : out unsigned(3 downto 0) -- פלט (4 ביטים)
    ;
;end ALU

architecture Behavioral of ALU is
    begin
        process(A, B, Opcode)
            begin
                case Opcode is
                    when "00" => -- A + B
;Y <= std_logic_vector(resize(unsigned(A) + unsigned(B), 4))
                    when "01" => -- A and B
```

;Y <= A and B

when "10" => -- A or B

;Y <= A or B

when "11" => -- A xor B

;Y <= A xor B

<= when others

(Y <= (others => '0')) -- ברירת מחדל: אפסים

;end case

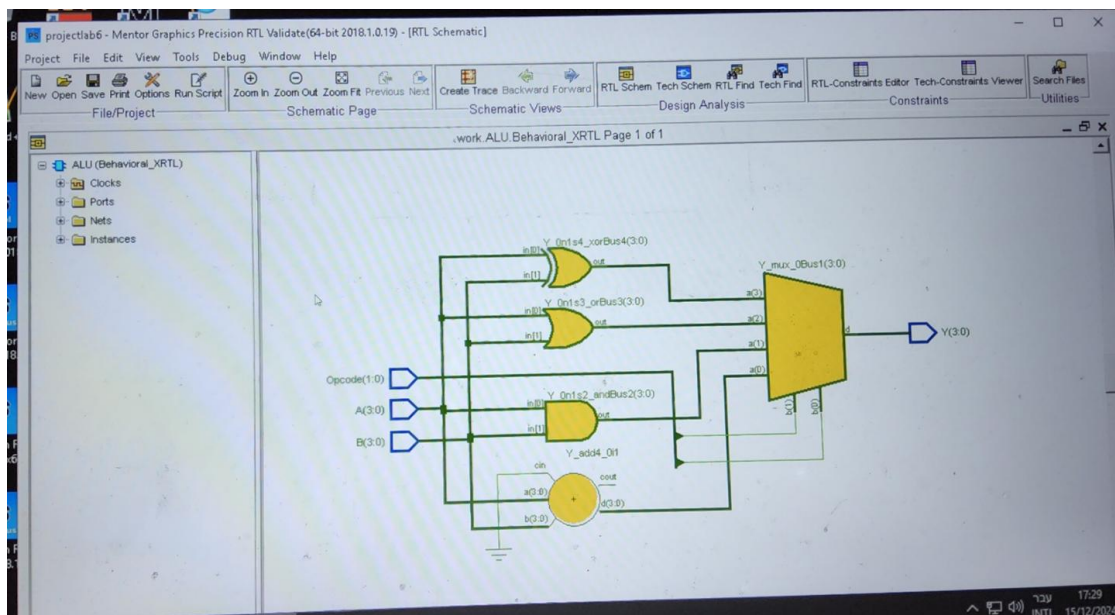
;end process

;end Behavioral

חלק ב, סינתזות

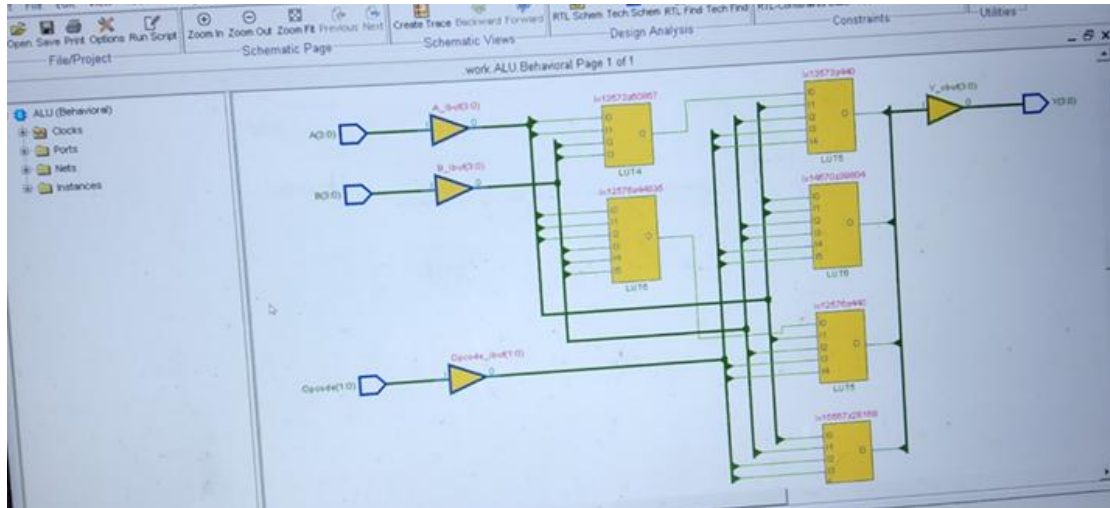
סעיף א

rtl



סעיף ב

סכימה של הטכנולוגיה



סעיף ג

Area report

Precision RTL Validate 64-bit 2018.1.0.19 (Production Release) Fri Aug 10 05:55:53 PDT // 2018

//

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//

Running on Windows Server 2016 ybrim@MSTS-ELEC 10.0.17763 x64 //

//

Start time Sun Dec 15 17:25:13 2024 //

Device Utilization for 7A15TCPG236

Resource	Used	Avail	Utilization

IOs	14	106	13.21%

Global Buffers	0	32	0.00%
LUTs	6	10400	0.06%
CLB Slices	2	8150	0.02%
Dffs or Latches	0	20800	0.00%
Block RAMs	0	25	0.00%
DSP48E1s	0	45	0.00%

Library: work Cell: ALU View: Behavioral

Cell Library References Total Area

	IBUF	xca7	10 x
LUT4	xca7	2 x	1 2 LUTs
LUT5	xca7	2 x	1 2 LUTs
LUT6	xca7	2 x	1 2 LUTs
	OBUF	xca7	4 x

Number of ports : 14

Number of nets : 30

Number of instances : 20

Number of references to this view : 0

: Total accumulated area

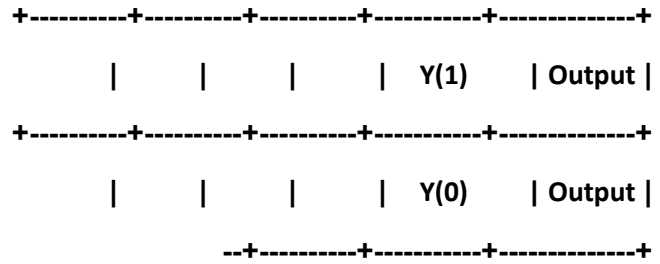
Number of LUTs : 6

Number of accumulated instances : 20

IO Register Mapping Report

Design: work.ALU.Behavioral

Port	Direction	INFF	OUTFF	TRIFF
			A(3)	Input
			A(2)	Input
			A(1)	Input
			A(0)	Input
			B(3)	Input
			B(2)	Input
			B(1)	Input
			B(0)	Input
			Opcode(1)	Input
			Opcode(0)	Input
			Y(3)	Output
			Y(2)	Output



סעיף ד

Timing report

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Running on Windows Server 2016 ybrim@MSTS-ELEC 10.0.17763 x64 //

//

Start time Sun Dec 15 17:25:13 2024 //

Device: Xilinx - ARTIX-7 : 7A15TCPG236 : 3 --

..CTE report summary --

POST-SYNTHESIS TIMING REPORTS ARE ESTIMATES AND SHOULD NOT BE RELIED ON TO --

MAKE QoR DECISIONS. For accurate timing information, please run place-and-route (P&R)
and review P&R generated timing reports

=====
=====

Clock Frequency Report

Domain	Clock Name	Min Period (Freq)	Required Period (Freq)
-----	-----	-----	-----
VirtualClock_PS	VirtualClock	4.758 (210.172 MHz)	10.000 (100.000 MHz)

=====

=====

Setup Timing Analysis of VirtualClock

Setup Slack Path Summary

Data			Data			Data		
Setup Path			Data			End	End	
Index	Slack	Delay	Source	Clock	Dest. Clock	Start Pin	Pin	Edge
-----	---	---	-----	-----	-----	---	---	---
	VirtualClock	VirtualClock	A(0)	Y(3)	Rise	4.758	5.242	1
	VirtualClock	VirtualClock	B(0)	Y(3)	Rise	4.758	5.242	2
	VirtualClock	VirtualClock	A(1)	Y(3)	Rise	4.747	5.253	3
	VirtualClock	VirtualClock	B(1)	Y(3)	Rise	4.747	5.253	4
	VirtualClock	VirtualClock	A(2)	Y(3)	Rise	4.742	5.258	5
	VirtualClock	VirtualClock	B(2)	Y(3)	Rise	4.742	5.258	6
	VirtualClock	VirtualClock	Opcode(0)	Y(3)	Rise	4.366	5.634	7
	VirtualClock	VirtualClock	Opcode(1)	Y(3)	Rise	4.366	5.634	8
	VirtualClock	VirtualClock	A(3)	Y(3)	Rise	4.348	5.652	9
	VirtualClock	VirtualClock	B(3)	Y(3)	Rise	4.348	5.652	10

CTE Path Report

:Critical path #1, (path slack = 5.242)

SOURCE CLOCK: name: VirtualClock period: 10.000000

Times are relative to the 1st rising edge

DEST CLOCK: name: VirtualClock period: 10.000000

Times are relative to the 2nd rising edge

NAME	GATE	DELAY	ARRIVAL	DIR	FANOUT
	A(0)	(port)		0.000	dn
	A(0)	(net)	0.000		1
	A_ibuf(0)/I	IBUF		0.000	dn
	A_ibuf(0)/O	IBUF	1.147	1.147	dn
	A_int(0)	(net)	0.313		4
	ix12576z44835/I2	LUT6		1.460	dn
	ix12576z44835/O	LUT6	0.097	1.557	dn
	nx12576z2	(net)	0.295		1
	ix12576z440/I0	LUT5		1.852	dn
	ix12576z440/O	LUT5	0.097	1.949	up
	nx12576z1	(net)	0.295		1
	Y_obuf(3)/I	OBUF		2.244	up
	Y_obuf(3)/O	OBUF	2.514	4.758	up
	Y(3)	(net)	0.000		0
	Y(3)	(port)		4.758	up

Initial edge separation: 10.000

Source clock delay: - 0.000

Dest clock delay: + 0.000

Edge separation: 10.000

Setup constraint: - 0.000

Data required time: 10.000

Data arrival time: - 4.758 (81.02% cell delay, 18.98% net delay)

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סעיף ה

סינטזה

synthesize

.Info: [662]: Current working directory: Z:/vhdlab6/projectlab6_impl_1 #

Warning: [19539]: No debug point is implemented, continuing with the normal #
.synthesis run

Info: [15002]: Optimizing design <...>.ALU.Behavioral #

Info: [15002]: Optimizing design <...>.ALU.Behavioral #

...Info: [8010]: Gated clock transformations: Begin #

...Info: [8010]: Gated clock transformations: End #

Info: [1081]: Constraints: Precision will generate the following clock domain information #
against the clocks. Since the clock domain information is not provided by the user for the
below clocks, all clocks are assumed to be synchronous clocks. This will result in all paths
between the clock domains to be analyzed for timing. If this behaviour is not intended for
any clock domain or path, please override using -domain option or false path constraint.
Among these clocks, clock with the smallest period will be chosen as the default domain
.and the rest of the clock periods specified using multiply-by or divide-by factor

:Info: Constraints: Precision Generated Clock Domains #

Info: Domain Name: VirtualClock_PS Clock Name: VirtualClock #

.Info: [3027]: Writing file: <...>/projectlab6_impl_1/ALU.edf #

Info: -- Writing file Z:/vhdlab6/projectlab6_impl_1/ALU.tcl #

.Info: [3027]: Writing file: <...>/projectlab6_impl_1/ALU.xdc #

.Info: [3027]: Writing file: <...>/projectlab6_impl_1/ALU.v #

Info: -- Writing file Z:/vhdlab6/projectlab6_impl_1/ALU.tcl #

...Info: [12045]: Starting timing reports generation #

.Info: [12046]: Timing reports generation done #

Info: [12048]: POST-SYNTHESIS TIMING REPORTS ARE ESTIMATES AND SHOULD NOT BE #
RELIED ON TO MAKE QoR DECISIONS. For accurate timing information, please run place-
.and-route (P&R) and review P&R generated timing reports

.Info: [665]: Finished synthesizing design #

.Info: [11019]: Total CPU time for synthesis: 0.7 s secs #

.Info: [11020]: Overall running time for synthesis: 1.3 s secs #